SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I D. Dharshana (192524243) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Dharshana

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CSA05111 - CRYPTOGRAPHY & NETWORK SECURITY

CO1 AT-1 - Application Based Questions

NAME : D. DHARSHANA
REG NO : 192524243

scenario : Event communication Security

A Student organisation is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Caesar Cipher.

The caesar is classical substitution encryption technique in which each letter in the plain text is shifted by a fixed number of position in alphabet. In this scenario, the Message "MEET AT NOON" is encrypted using a shift of 4 positions.

a) Encrypt the message using a shift of 4 positions.

Plain text : MEET AT NOON
Shift key : K:4
Encryption Method : Caesar cipher

$$C = (P + K) \text{ Mod } 26$$

C → cipher text letter value
P → plain text letter value
K → shift key.

A	B	C	D	E	F	G	H	I	J	K	L
0	1	2	3	4	5	6	7	8	9	10	11

M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
12	13	14	15	16	17	18	19	20	21	22	23	24	25

MEET	AT	NOON
12 4 4 19	0 19	13 14 14 13
| | | |	| |	| | | |
Q I I X	E X	R S S R

$C(M) = (12 + 4) \% 26$
$= 16 \% 26 = 16 = Q.$

$C(E) = (4 + 4) \% 26$
$= 8 \% 26 = 8 = I$

$C(T) = (19 + 4) \% 26$
$= 23 \% 26 = 23 = X$

$C(A) = (0 + 4) \% 26$
$= 4 \% 26 = 4 = E$

$C(N) = (13 + 4) \% 26.$
$= 17 \% 26 = 17 = R$

$C(O) = (14 + 4) \% 26$
$= 18 \% 26 = 18 = S$

Cipher text = Q IIX EX RSSR.

ow an authorized member decrypts the message.

An authorized member who knows the shift key (A) can ecrypt the cipher text by shifting each letter four positions backward in the alphabet.

Decryption formula :

$$P = (C - K) \% 26$$

16 8 8 23 4 23 17 18 18 17
(Q I I X E X R S S R)
M E E T A T N O O N

$P(Q) = (16 - 4) \% 26$
$= 12 \% 26 = 12 = M$

$P(I) = (8 - 4) \% 26$
$= 4 \% 26 = 4 = E$

$P(X) = (23 - 4) \% 26$
$= 19 \% 26 = 19 = T$

$P(E) = (4 - 4) \% 26$
$= 0 \% 26 = 0 = A$

$P(R) = (17 - 4) \% 26$
$= 13 \% 26 = 13 = N$

$P(S) = (18 - 4) \% 26$
$= 14 \% 26 = 14 = O$

Plain Text : MEET AT NOON

Advantages.

1. Simple to understand + implement.
2. Require only fixed shift key.
3. Useful for learning basic encryption + drayption concepts.

Limitations:

1. It has only 25 possible non-zero shift key, brute-force attacks easy.
2. Letter frequency pattern remain unchanged.
3. It is not suitable for protecting highly confidential information in modern communication systems.

2 Scenario: Intercepted Suspicious Message

A cybersecurity intern the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures or known cipher characteristics.

a) Decryption:

The intercepted ciphertext is "GSRH RH Z HVXIVG"

This message can be decrypted using monoalphabetic substitution cipher. The reverse-alphabet substitution method.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M
O	P	Q	R	S	T	U	V	W	X	Y	Z		
L	K	J	I	H	G	F	E	D	C	B	A		

3.

GSRH	RH	Z	HVXIVG
THIS	IS	A	SECRET

Plain text : THIS IS A SECRET

Justification

1. Simple Substitution Mechanism: The ciphertext contains uppercase letters + spaces, suggesting that word boundaries have been preserved. This is consistent with a classical substitution cipher.

2. Letter-by-Letter replacement: Every letter is replaced by its reverse alphabet counterpart. For example, G → T, S → H R → I + H → S. Therefore, GSRH → THIS

3. Word Structure: The cipher text contains 4 words with length 4, 2, 1, 6. The decrypted message, THIS IS A SECRET has exactly the same-word length pattern.

4. No Secret key required: It uses a fixed alphabet-reversal rule, so sender + receiver do not need to exchange a seperations encryption key.

3. Scenario : Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.

b) Explain how the repeating keyword influences each character of ciphertext.

a) Encryption process Step-by-Step.

The Vigenere cipher is a polyalphabetic substitution cipher that uses a repeating keyword to encrypt a plain text message.

Given:

* Plain Text : HELLO
* key word : KEY
* Encryption Method : Vigenere cipher

$$C = (P + K) \% 26$$

$$C(H) = (H + K) \% 26$$
$$= (7 + 10) \% 26$$
$$= 17 \% 26 = 17 = R$$

$$C(E) = (4 + 4) \% 26$$
$$= 8 \% 26 = 8 = I$$

$$C(L) = (11 + 24) \% 26$$
$$= 35 \% 26 = 9 = J$$

$$C(L) = (11 + 10) \% 26$$
$$= 21 \% 26 = 21 = V$$

$$C(O) = (14 + 4) \% 26$$
$$= 18 \% 26 = 18 = S$$

Cipher text : R I J V S

$$\begin{matrix} T & H & E & L & L & O \\ H & E & L & L & O & \\ K & E & Y & K & E & \\ 10 & 4 & 24 & 10 & 4 & \end{matrix} \%$$

H E L L O
R I J V S

the repeating keyword influences each character.

Keyword Repetition : The keyword KEY is repeated as KEYKE to match the length of the plain text.

Different Shifts : Each keyword letter determine the shift applied to its corresponding plaintext letter. For example, K shifts H by 10 position, while E shifts E by 4 position.

3. Multiple substitution alphabets: Unlike the caesar cipher, which uses 1 fixed shift, Vigenere cipher uses different shift based on the keyword letter.

4. Improved Security : The repeating keyword makes the encryption pattern less obvious than a simple caesar cipher. although short or repeated keywords can still be vulnerable to cryptanalysis.

5. Decryption : The receiver uses the same keyword & subtract the key value from each cipher text letter using :

$$P = (C - K) \text{ Mod } 26.$$

This recovers the original message HELLO.

4. Scenario : Suspicious Digital Asset Investigation.

During a digital forensic audit. Investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a) Analyse how investigation could detect the presence of hidden data within the image.

b) Recommend two techniques that ... enhance the security of such hidden communications against the detection & extraction.

a) Methods to detect hidden data in an image.

IMAGE STEGANOGRAPHY is a technique used to hide confidential information inside an image file without visible changing its appearance. Investigators can use the following methods to detect hidden data.

1. Visual Inspection: Compare the original image with suspected image & examine unusual colours, pixels or patterns. However data may not be visible to naked eye.

2. File size analysis: check whether the image file size is unusually large compared with similar image of same resolution & format: Extra hidden information may increase the file size.

3. Metadata analysis: Examine metadata such as file format, dimensions, Machine setup & connects using forensic tools. Unusual metadata may indicate suspicious activity.

4. Statistical Analysis: Analyze pixel values, color distribution & patterns. Stegnography may introduce statistical irregularities that suggest hidden information.

5. Stegnography tools: Use forensic tools such as stegexpose or stegdetect, where applicable, to identify suspicious patterns associated with certain stegnographic methods.

technique to strengthen the security of hidden communications

Encryption before embedding.

Encrypt the confidential message using a strong encryption algorithm such as AES before embedding it into the image.

→ Even if investigations extract the hidden data. they cannot easily understand it without decryption key.

→ It prodes an additional layer of confidentiality.

→ It protects the message even if the stegnography method is discovered.

2. Least significant Bit (LSB) stegnography with randomized embedding.

Embeded encrypted data in selected pixel bits using a secret. Pseudorandom embedding pattern rather than storing it sequentially

→ It can make hidden data harder to locate through simple visual inspection.

→ Randomized positions can reduce obvious patterns associated with sequential embedding.

→ The receiver needs the correct secret key or pattern to locate + extract the message.

5) Scenario: Legacy Encryption in corporate communication.

A legacy system in a company still uses a transposition -based encryption method with three levels of arrangement. A received message appears as:

"WECRLTEFRDSOEFFEAOCAIVDEN".

a) Determine the most probable original message by recovering the encryption process.

b) Initially evaluate why such a method is vulnerable in modern communication systems.

c) Propose one practical enhancement to improve the confidentiality of message in this scenario.

a) Decryption:

Given cipher text: "WECRLTEERDSOEEFEADCAIVDEN"

The Rail Fence cipher is a transposition cipher that rearranges letters in a zigzag pattern across three rule. The exact plaintext cannot be confirmed without verifying the encryption pattern & rail arrangement.

Decrypt Plaintext: WE ARE DISCOVERED FLEE AT ONCE using a 3-Rail Fence cipher

b) Vulnerabilities

1. Small key spaces: Easy to break by trying different rail arrangements.

2. Letter frequency: Original Letter Frequencies.

Weak Security : Does not provide sufficient confidentiality against modern attack.

One practical enhancement.

Use AFS . encryption before transmission.

The company should replace the legacy transposition -only method with AES -256 . a modern symmetric encryption algorithm. The sender encrypts the original message using a secret key + the authorized receiver decrypts it using the corresponding key.

advantages :

-> provides strong confidentiality.

-> Makes unauthorized decryption computationally impractical when implemented completely.

-> can be used with ~~authorization~~ authenticated encryption mode such AFS-GCM to detect tampering.